# Semantic Trajectory Analysis on Canonical v2 Outputs

Weighted semantic links across meetings, using the updated v2 annotation schema

# Why This Layer

- The canonical v2 outputs already track idea trajectory, decision crystallization, and related meeting-state signals.
- This layer uses a fuzzy-linkography approach to add weighted links between utterances across the meeting.
- The current implementation uses Latent Semantic Analysis as the semantic-similarity model inside that approach.
- The current rerun also filters out auxiliary ATTN files.

# Coverage

| Metric | Value |
|---|---|
| Conferences included | 8 (2020NES, 2021ABI, 2021CMC, 2021MND, 2021MZT, 2021NES, 2021SLU, 2022MND) |
| Meetings analyzed | 179 |
| Chunks analyzed | 1219 |
| Sessions with outcomes | 136 |
| Methodology | Fuzzy linkography |
| Similarity model | Latent Semantic Analysis |

# What Comes From The JSONs vs What

| Directly from Evey's JSON outputs | Added by fuzzy linkography |
|---|---|
| chunk_summary.idea_trajectory | weighted links inferred between utterances across the full meeting |
| chunk_summary.decision_crystallization_level | meeting-level summary scores built from those inferred links |
| chunk_summary.collective_engagement_level | cross-chunk connection structure across the whole session |
| utterance_annotations | a session-level layer that is not directly stored in the JSON files |

* The JSON files already contain chunk-level annotations. The fuzzy-linkography step adds a separate meeting-level layer by inferring weighted semantic links across utterances over time.

# Worked Example: Added Meeting-Level

Conference: 2020NES | Session: 2020_11_05_NES_S1 | Meeting: 1_DAC_Simulations_Zoom_Meeting_2020_11_05_12_24_10

| Chunk | Time | idea_trajectory | decision level | full utterance quote |
|---|---|---|---|---|
| 1 | 00:00 | divergent | 1 | The main question we are asked in this session is how your new technology development or new materials would interact with the earth system basically. One of the reasons people like converting CO2 to high energy density hydrocarbon is we already know how to move hydrocarbon all over the globe without losing energy density too much. how your new technology development |
| 3 | 00:04 | divergent | 1 | I definitely think so Valentina that we should have a set of... wide range of technologies. direct air capture in Beijing versus New York would be very different |
| 5 | 00:02 | convergent | 3 | if you look at CO2 hydrogenation to something like methanol, you need high pressures... being at sub-atmospheric pressures really hurts you in terms of equilibrium conversions. there needs to be a significant energy input in the form of pressure... and you need a source of hydrogen. based on my reading... you need high pressures |

| From | To | Weight |
|---|---|---|
| Chunk 1 | Chunk 5 | 0.997 |

* Worked example from one canonical SCIALOG meeting. Decision level comes from chunk_summary (1 = earlier/ open discussion, 3 = more crystallized discussion). The left table comes directly from the JSON content; the right table is added by fuzzy linkography, which infers weighted links between those utterances across the meeting. Each row is one utterance from a longer meeting dialogue, not the full dialogue.

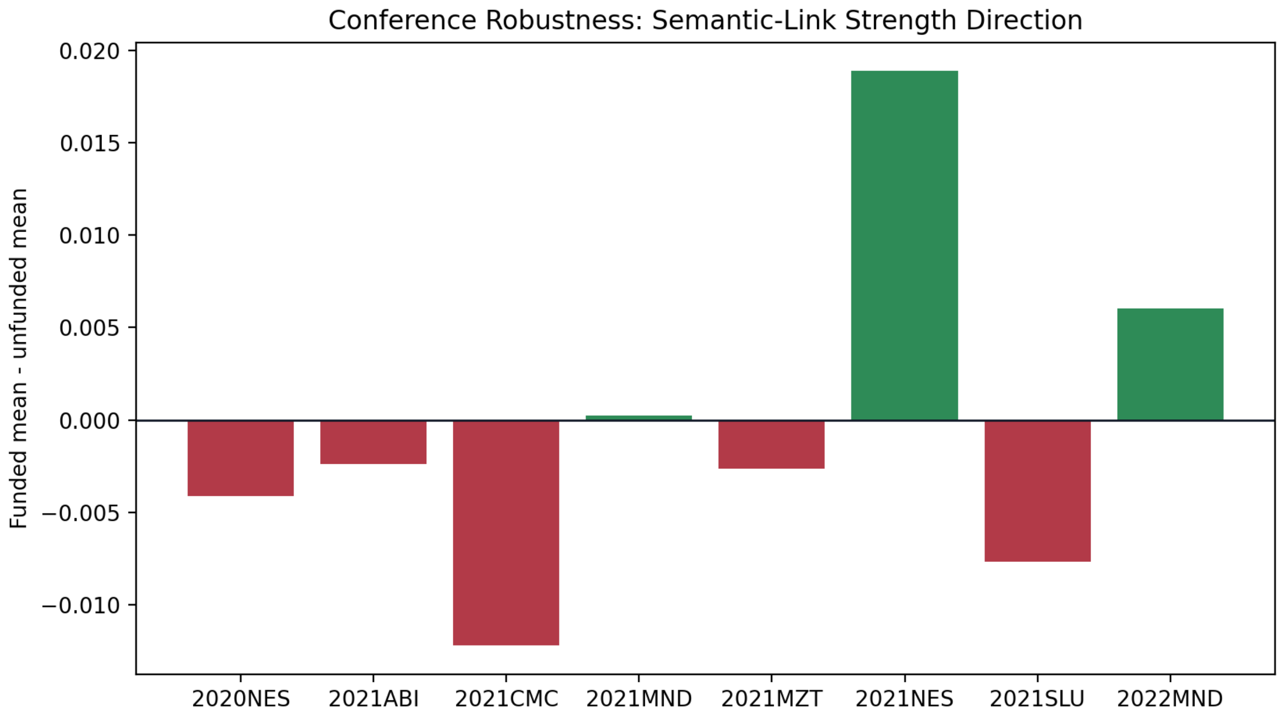
# Current Direct Comparison

- When the goal is funding, this added meeting-level layer does not give one clear score that separates funded from unfunded sessions.
- Some differences appear in individual comparisons, but they do not stay consistent enough to treat as the main finding.
- So the main thing this analysis is pulling out right now is session-level structure, not a clean funding signal.

# How This Connects To The v2 JSON Terms

| v2 JSON term | What it already gives us | What the added meeting-level layer contributes |
|---|---|---|
| idea_trajectory | The chunk-level direction of discussion in the JSONs | This added layer looks across the whole meeting and asks how strongly utterances stay related over time |
| decision_crystallization_level | How clearly the discussion is moving toward a decision | This added layer tracks whether later discussion reconnects to earlier content as decisions develop |
| collective_engagement_level | How active and mutually engaged the group seems | This added layer can show whether continuity is mostly within one speaker or carried across speakers |
| shared_vision / commitment flags | Signals that the team is aligning around an idea | This added layer helps show whether that alignment is sustained across the meeting rather than only inside one chunk |

# Across Conferences




One direction repeats in 5 of 8 conferences, but not across the full corpus.
Direction is mixed rather than stable across cohorts.
So this does not support one robust conference-invariant funding story.

# Conference Breakdown

| Conference | N | Funded Mean | Unfunded Mean | Diff |
|---|---|---|---|---|
| 2020NES | 16 | 0.177 | 0.181 | -0.004 |
| 2021ABI | 18 | 0.186 | 0.189 | -0.002 |
| 2021CMC | 18 | 0.192 | 0.204 | -0.012 |
| 2021MND | 23 | 0.181 | 0.181 | +0.000 |
| 2021MZT | 15 | 0.171 | 0.174 | -0.003 |
| 2021NES | 15 | 0.199 | 0.180 | +0.019 |
| 2021SLU | 20 | 0.168 | 0.176 | -0.008 |
| 2022MND | 11 | 0.181 | 0.175 | +0.006 |

# Example Meetings To Review: Higher-Score Cases

| Conference | Session | Chunks | Funded |
|---|---|---|---|
| 2020NES | 2020_11_05_NES_S1 | 2 | 0 |
| 2021MND | 2021_04_23_MND_S1 | 2 | 0 |
| 2021MND | 2021_04_22_MND_S8 | 1 | 0 |
| 2021NES | 2021_11_04_NES_S6 | 4 | 1 |

* Review examples only: these meetings were selected from the highest and lowest mean_nonzero_weight values after filtering to meetings with at least 10 extracted moves. They are not 'best' or 'worst' meetings.

# Example Meetings To Review: Lower-Score Cases

| Conference | Session | Chunks | Funded |
|---|---|---|---|
| 2020NES | 2020_11_05_NES_S6 | 2 | 0 |
| 2021MND | 2021_04_22_MND_S7 | 2 | 0 |
| 2020NES | 2020_11_05_NES_S5 | 2 | 1 |
| 2020NES | 2020_11_06_NES_S1 | 4 | 1 |

* Review examples only: these meetings were selected from the highest and lowest mean_nonzero_weight values after filtering to meetings with at least 10 extracted moves. They are not 'best' or 'worst' meetings.